



SHZVG VACUUM ON-LOAD TAP-CHANGER



Quality Creates Value

Shanghai Huaming Power Equipment Co., Ltd.

PRODUCT CHARACTERISTICS



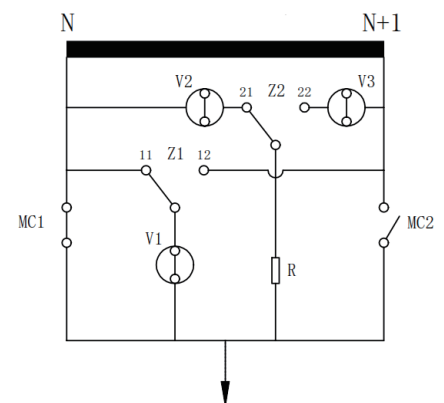
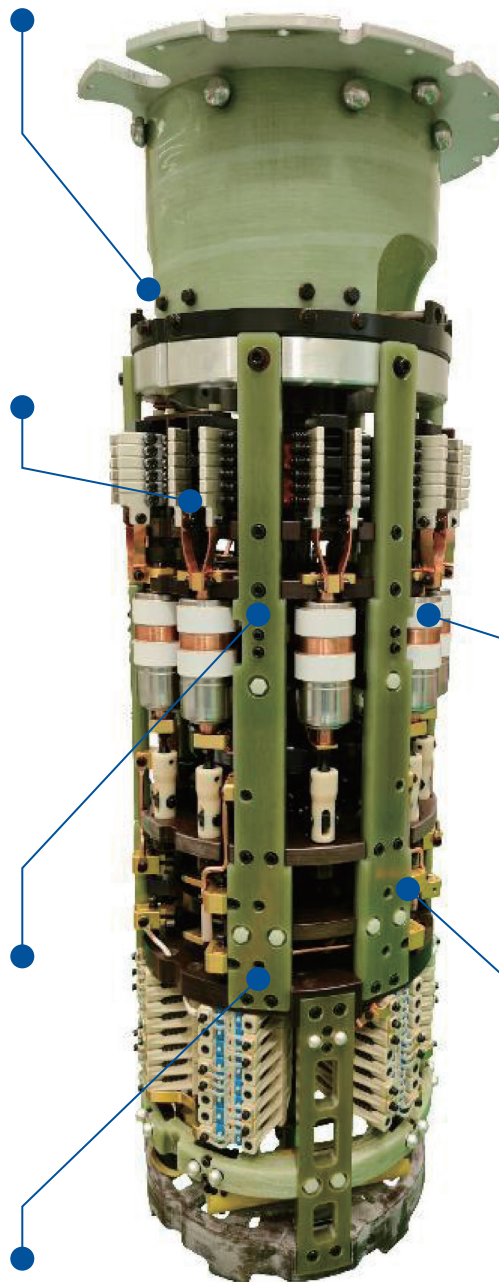
The SHZVG oil-immersed vacuum on-load tap-changer, with a maximum rated step voltage of 6000V, a maximum rated current of 1500A, and a maximum step capacity of 6000 kVA, is suitable for a variety of converter transformers and flexible DC transformers in ultra-high voltage DC (UHVDC) systems, as well as other high-capacity applications.

It features a stable and reliable high-angle output, rapid-acting mechanism that provides ample switching time. The cast steel base significantly enhances structural strength.

MC: The "multi-piece" contact design offers high current-carrying capacity and has passed the 20kA/3s thermal stability test, exceeding the national standard duration by 50%.

Z1: "Clamshell" isolating contacts eliminate energized sliding and metal debris generation.

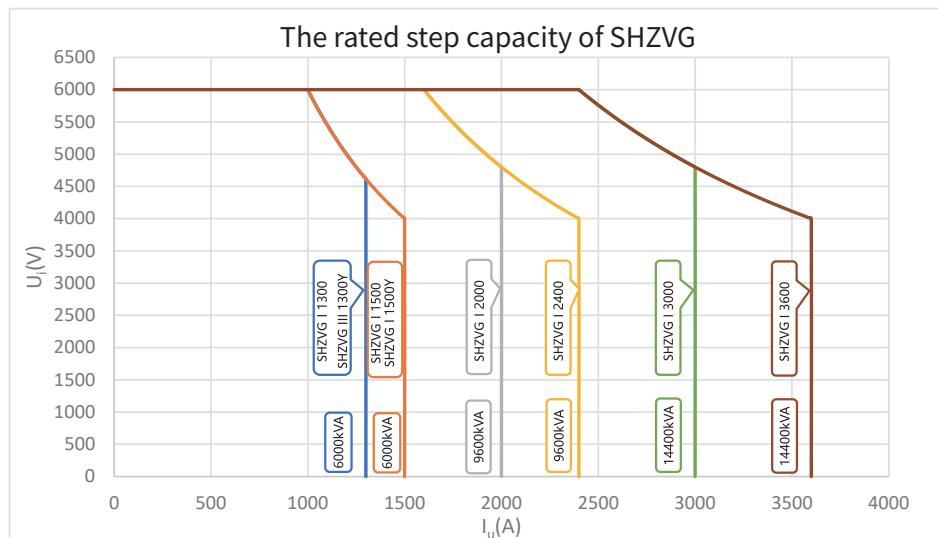
Modular design with "snap-fit" installation ensures high structural integrity.



Electrical Topology Diagram

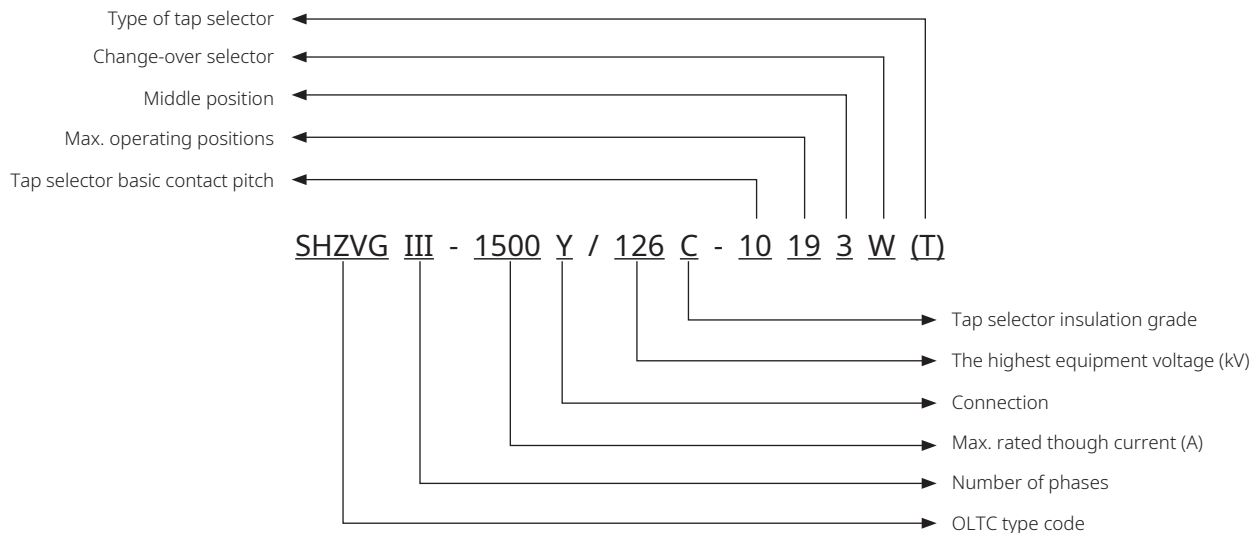
3. The vacuum redundancy design, with alternating load switching, reduces vacuum interrupter operating frequency and extends their service life.

Z2: The quasi-symmetrical mechanism actuates before the end of each switching operation, consuming no additional switching time and providing a large operational margin.



The rated step capacity is defined as the product of the step voltage and the load current, i.e. $PS = UI$. The rated step capacity is the maximum step capacity of the on-load tap-changer under continuous operating conditions, i.e. $P_{STN} = I_{ux} \times U_i$.

Code description



Conditions

1. The operating temperature of the tap-changer in oil is -25°C to +105°C.
2. The temperature of the ambient environment where the on-load tap-changer is to be used is - 25 °C to + 40 °C.
3. The perpendicularity of the on-load tap-changer to the ground after mounting on the transformer should not exceed 2%.
4. The place of the on-load tap-changer located must be free of serious dust and other explosive or corrosive gases.

Note: For more severe environmental conditions of the on-load tap-changer or motor-drive unit, please contact with Huaming technical department for further confirmation.

Applicable Standards: IEC60214-1: 2014 and GB10230.1-2019

Type		SHZVG III 1300Y SHZVG I 1300	SHZVG III 1500Y SHZVG I 1500	SHZVG I 2000	SHZVG I 2400	SHZVG I 3000	SHZVG I 3600
Max. Rated Through Current (A)		1300	1500	2000	2400	3000	3600
Rated Frequency (Hz)		50 or 60					
Connection		Star Connection at Neutral Single-phase for any connection					
Max. Rated Step Voltage (V)		6000					
Rated Step Capacity (kVA)		6000		9600		14400	
Short circuit capacity (kA)	Thermal(3s)	20		30		40	
	Dynamic (peak)	50		75		100	
Max. operating positions		18 without change-over selector; 35 with change-over selector					
Insulation level to earth (kV)	The highest voltage for equipment (kV)	72.5	126	170	252	300	363
	Rated separate source AC withstand voltage (kV/50Hz, 1min)	140	230	360	460	480	510
	Rated lightning impulse withstand voltage (1.2/50μs)	350	550	750	1050	1100	1175
Tap selector		Categorized into B, C, D, DE four sizes					
Mechanical life		1,500,000					
Maintenance free		300,000					
Diverter switch oil compartment	Service pressure	0.03MPa					
	Leakage test	0.1MPa 24h/ No leakage					
	Over pressure protection	Rupture disc bursts at 0.3±20%MPa					
	Protective relay	2.5m/s±10%					
Motor drive unit		SHM- Series or CMA7					